

HIGH-PERFORMANCE LIQUID CHROMATOGRAPHY
(HPLC) & BETA-GLOBIN GENETICS

Date: September 08, 2026

Patient Name:	Kavita Sundaram	MRN / Patient ID:	CMC - LUD - THAL - 4491
Age / Gender:	8 Yrs / Female	Blood Group & Rh:	B+ (B Rh Positive)
Clinical Diagnosis:	Transfusion-Dependent Beta Thalassemia Major	Referring Consultant:	Dr. Joseph John, MD, DM

Hemoglobin Fraction	Observed Percentage	Normal Biological Reference	Diagnostic Significance
Hemoglobin F (HbF)	88.5%	< 2.0% in adults	Marked Elevation / Severe Thalassemia
Hemoglobin A0 (HbA)	9.4%	95.0% - 98.0%	Severe Suppression of Beta Chains
Hemoglobin A2 (HbA2)	2.1%	2.0% - 3.5%	Normal Level
Beta-Globin Gene Mutation	IVS 1-5 (G>C) Homozygous	Normal Wild-Type	Classic Severe Indian Mutation
Serum Ferritin Level	2,840 ng/mL	15 - 150 ng/mL	Significant Secondary Iron Overload

CLINICAL INTERPRETATION & RECOMMENDATIONS:

- High-performance liquid chromatography and Sanger sequencing demonstrate homozygous IVS 1-5 mutation.
- Findings confirm severe Transfusion-Dependent Beta Thalassemia Major requiring chronic iron chelation.
- Recommendation: Curative allogeneic stem cell transplantation is indicated. Sibling and MUD registry search prioritized.

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